

26 November 2020

Number:

Name Surname:

**CEN 209 Differential Equations
QUESTIONS**

- 1.** Find the general solution of the differential equation

$$(2y \sin x \cos x + y^2 \sin x)dx + (\sin^2 x - 2y \cos x)dy = 0$$

- 2.** Find the general solution of the differential equation

$$\cos y dx + (x \sin y - \cos^4 y)dy = 0$$

- 3.** Find the general solution of the differential equation

$$y^3 dx + (x^3 - xy^2)dy = 0$$

- 4.** Find the general solution of the differential equation

$$x \frac{dy}{dx} + y = 2x^6 y^4$$

- 5.** Find the general solution of the differential equation

$$(xy^2 + y^2 + y)dx + (2xy + 1)dy = 0$$

Time: 75 minutes.

Best Luck